A KNOWLEDGE-ENHANCED CHATBOT SYSTEM USING VERIFIED REVIEW INTELLIGENCE FOR E-COMMERCE PLATFORMS

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this report, titled "**A Knowledge-Enhanced Chatbot System Using Verified Review Intelligence for E-commerce platforms**" is the bonafide work of **Lokesh Adithya B (3122215002053), D Sachin Abhishek (3122215002088) and Rupam Banik (3122215002086)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

Customers generally find it hard to differentiate between real and fake reviews of products because e-commerce has grown rapidly. Shopping choices can be significantly affected by such difficulty, and hence an authentic review verification system is needed. In a bid to determine and mark off fake reviews, our project introduces an advanced review verifying mechanism powered by machine learning and natural language processing (NLP). Our approach assures greater transparency and trust in e-commerce product reviews using authentic review intelligence.

With the Phi-3 enhanced model, we have also developed an intelligent chatbot that assists users in finding product reviews more effectively. To give users contextual responses to queries regarding the performance, usability, and durability of products, the chatbot extracts, summarizes, and analyzes verified reviews. By eliminating the necessity for users to sift through numerous reviews manually, this automated review analysis speeds up the decision-making process.

By means of real-time learning mechanisms and user feedback loops, our system constantly adapts to evolving deceptive methods, ensuring long-term effectiveness. We enhance the user experience and provide a simple-to-use interface for interaction with verified product reviews by incorporating a robust frontend application. Both customers and ethical sellers benefit from this solution's creation of a more trustworthy e-commerce environment.

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TABLE OF CONTENTS

CHAPTER NO	TITLE	PAGE NO
	ABSTRACT	3
	LIST OF TABLES	8
	LIST OF FIGURES	9
	LIST OF ABBREVIATION	12
1	**INTRODUCTION**	13
2	**LITERATURE SURVEY**	19
3	**DATASET DESCRIPTION**	26
	3.1. IDENTIFIERS AND METADATA	26
	3.2. RATINGS AND VOTES	27
	3.3. TEXTUAL DATA	27

	3.4. VERIFICATION AND MODERATION	28
4	**SYSTEM DESIGN AND ARCHITECTURE**	29
5	**MODEL TRAINING**	33
	5.1. FEATURE SELECTION	33
	5.2. TEXT FEATURE TRANSFORMATION USING BERT EMBEDDINGS	33
	5.3. NUMERICAL FEATURE PROCESSING	34
	5.4. FEATURE COMBINATION	34
6	**EXPLAINABILITY**	35
	6.1. FEATURE VALUES	35
	6.2. INTERMEDIATE CALCULATIONS	35
	6.3. FEATURE WEIGHTS	35

	6.4. LIME EXPLANATIONS	36
	6.5. HUMANIZED EXPLANATION VIA PHI-3 OLLAMA CHATBOT	36
	6.6 FINAL DECISION	37
7	**EXPERIMENT AND RESULTS**	39
8	**EVALUATION**	41
9	**SOCIAL IMPACT**	49
10	**CONCLUSION**	52
11	**LIMITATIONS AND FUTURE WORK**	54
12	**REFERENCES**	56

LIST OF TABLES

TABLE NO	DESCRIPTION	PAGE NO
Table 8.1	Ablation Study Results Comparing Different Models for Fake Review Detection	41
Table 8.2	Evaluation table for different LLM'S and SLM's.	44
Table 8.3	Showing Benchmark for AI Chatbots.	45

LIST OF FIGURES

FIGURE NO	DESCRIPTION	PAGE NO
Figure 4.1	Showing Flow Diagram.	31
Figure 4.2	Showing Architecture Diagram.	32
Figure 7.1	Showing Application UI with an example review.	39
Figure 7.2	Showing predicted fake review with explainability.	39
Figure 7.3	Showing predicted genuine review with explainability.	39
Figure 7.4	Showing TrustScore chatbot interface.	40
Figure 7.5	Showing the dataset after classification.	40
Figure 8.1	Showing Training and Validation Accuracy for BERT-LSTM-CNN.	41
Figure 8.2	Showing Training and Validation Loss for BERT-LSTM-CNN.	41
Figure 8.3	Showing Effect of Helpness Score.	41

Figure 8.4	Showing model quality of different language models.	42
Figure 8.5	F1-score, precision, and recall metrics for our model with focal loss implementation	42
Figure 8.6	Distribution of confidence scores for genuine and fake reviews	42
Figure 8.7	ROC curve showing the performance of the hybrid model	43
Figure 8.8	Confusion matrix of the hybrid BERT-LSTM-CNN model with focal loss	43
Figure 8.9	Bar Chart showing Ease of Use of TrustScore.	45
Figure 8.10	Bar Chart showing Response Satisfaction of TrustScore.	46
Figure 8.11	Pie Chart showing Error Rate of TrustScore.	46
Figure 8.12	Pie Chart showing Response Speed of TrustScore.	46
Figure 8.13	Bar Chart showing Response Accuracy of TrustScore.	47
Figure 8.14	Pie Chart showing Task Completion Rate of TrustScore.	47

Figure 8.15	Bar Chart showing Speed and Efficiency of TrustScore.	47
Figure 8.16	Bar Chart showing Retention Rate of TrustScore.	48
Figure 8.17	Pie Chart showing performance comparison of TrustScore.	48

LIST OF ABBREVIATIONS

NO.	ACRONYM	ABBREVIATIONS
1	SVM	Support Vector Machines
2	NLP	Natural Language Processing
3	LSTM	Long Short-Term Memory
4	CNN	Convolutional Neural Network
5	RNN	Recurrent Neural Network
6	BERT	Bidirectional Encoder Representations from Transformers
7	LLM	Large Language Model
8	SLM	Small Language Model
9	RAG	Retrieval-Augmented Generation

1. INTRODUCTION

The rising number of online shopping platforms has significantly changed the way people shop, giving them accessibility to diverse category of products which wasn't available before. People were skeptical at first but the adoption period was fairly quick due to the convenience being offered. But however, this presented new challenges particularly with the integrity of product reviews. With e-commerce continuing to dominate the shopping landscape, shoppers are increasingly depending on reviews to make informed decisions while purchasing. According to studies, approximately 90% of the shoppers make their purchasing decisions based on the reviews of the product[1].

Sadly, the integrity of these reviews has been very questionable recently. The practice of posting reviews to - either boost sales by posting false positive reviews or to damage the competition by posting false negative reviews - has been on the rise and been a persistent issue across all online shopping platforms. These fabricated reviews mislead the shoppers while also undermining the authenticity of the whole e-commerce ecosystem. Studies indicate that upto 40% of the reviews online might be fabricated or manipulated[2], which reduces trust in the online shopping environment.

Beyond the challenge of fake reviews, consumers face an equally significant problem of information overload. Even among genuine reviews, many are irrelevant to specific purchasing decisions or focus

on aspects that may not matter to a particular consumer. Studies show that users typically read only 7-10 reviews before making a decision [3], highlighting the need for not just authentic but also relevant review content. This dual challenge—ensuring both authenticity and relevance—represents a critical gap in current e-commerce platforms.

The current review verification systems depend upon simple rule based systems or basic statistical models that often fail to catch the fine linguistic patterns which indicate fabricated reviews. These traditional methods often give high false positive rates and is unable to adapt to the highly changing fraudulent methods employed by the miscreants in the ecommerce marketplace. Further, the existing verification system doesnt address the issue of relevance of review, making consumers manually go through multiple genuine but potentially irrelevant reviews. NLP or natural language processing has become the foundation for transforming the e-commerce review verification systems. Our research continues on this path of transformation and provides some new concepts and approaches that help the transition from traditional verification systems to modern review verification systems and helps it adapt to the complex requirements of the current online platforms.

For this project, the most important input needed is a complete dataset containing artificial as well as genuine reviews which are seen in actual online marketplaces. To take it a step further, we need to make sure that

our system is equipped with intelligent and context aware review filtering mechanisms so that it is robust to changing tactics of bad actors. To satisfy this, our research takes upon the objective to develop a robust review verification system needed for this evolving digital marketplace. The paper understands the importance of text pre-processing in improving the quality of review analysis by extracting the necessary artifacts while discarding the unwanted ones and normalizing the text for further verification. This assures improvement in accuracy while recognition and classification of reviews irrespective of different patterns that may be followed. This forms the foundation for our model training.

Data augmentation methods help improve our model performance by enhancing our dataset. Augmentation procedures helps diversify our feature space significantly. These procedures include synonym replacement, swapping of words, random insertion and deletion. By this method, we prepare our machine learning models to adapt to various real- world challenges it might face and be ready to face them.

We aim to address the existing limitations in our research by developing a modern review verification system that integrates NLP techniques with deep learning architectures. Our project proposes a three-component neural network framework: a BERT (Bidirectional Encoder Representations from Transformers) model for semantic analysis, an LSTM (Long Short-Term Memory) network for sequential context

processing, and a CNN (Convolutional Neural Network) with variable filter sizes to identify and capture patterns across different lengths.

To address the significant class imbalance in our datasets - where we have observed that genuine reviews outnumber fabricated ones by a large margin - we have implemented a focal loss function that can dynamically adjust to pay attention to the examples which are generally difficult to classify. This allows us to increase the classification accuracy while training our model and and also maintains the computational efficiency.

Finally, we enhance the whole experience for the end user by implementing an intelligent chatbot using the Phi-3 small language model that is fed reviews from the verified relevant and genuine reviews' database. The chatbot while allowing users to directly access relevant genuine product reviews also provides insights extracted from the said reviews , summarizing it perfectly helping the user avoid wasting time going through the reviews manually to form a consensus about the product.

In the proposed model, the Bidirectional Encoder Representations from Transformers (BERT) has been utilized to transform words into tokens and find semantic relationships. This BERT algorithm has been combined with LSTM and CNN models to act as a hybrid innovative algorithm for identifying authentic reviews of various lengths, complexities, and linguistic patterns with good precision and in a faster

manner. For proving the efficacy of the proposed model, the results are compared with other models like traditional machine learning classifiers, rule-based systems, and simpler neural networks. The comparison proved higher outcomes in terms of precision, recall, and F1-scores. This comparison clearly suggests that the hybrid proposed innovative model outperforms all other models in the task of detecting fake reviews. The proposed model also achieved excellent GPU performance over time. This indicates that the proposed model is really suitable for real-time operation. One of the major strengths of the proposed model is its operation under varied review styles, product categories, and complex linguistic environments.

The major contributions of the research are detailed below:

- Development of a comprehensive review dataset enriched with linguistic features for improved classification.
- Improving dataset quality and diversity by undergoing several augmentation methods and preprocessing techniques, making our model robust to different styles of reviews.
- Improved accuracy for classification is achieved by integrating BERT, LSTM and CNN
- Addition of focal loss function to combat the class imbalance issues in our review datasets.
- Implementation of a chatbot using Phi-3 SLM model that provides users with summarized insights from verified relevant reviews.

- Presentation of a model superior in performance while also being efficient in real time, making the model suitable for the online marketplace, where robust , quick review verification might be needed.

These contributions collectively advance the state of review verification systems, offering a new framework that improves accuracy, efficiency, and especially the user experience in online shopping environments.

2. LITERATURE SURVEY

Review verification systems have been vastly covered by existing research, however there exist a few gaps in existing literature that have been bridged through our research. The literature review goes over present and past research as well as areas for improvement. It also points out our novelty and unique position in the review verification ecosystem [5]-[7].

Recent Advancements in Review Verification Systems:

Recent investigations under review verification systems focus majorly on NLP and machine learning techniques for improved system design and accuracy. Wang et al. explored a deep learning approach for real-time monitoring and filtering, thereby facilitating efficient fake review detection techniques[8],[9]. Another investigation by Liu demonstrated the working of these systems by providing accurate classification through advanced ML models [10],[11].

Another investigation in [12] demonstrates a comprehensive database with data that accurately encompasses all the tendencies of a fake as well as real data point.The models were optimized using deep learning approaches. The dataset comprises about 50,000 reviews across different categories completely annotated depicting realness and fakeness. Experiments with other state-of-the-art models show fairly good results which can be implemented in real world e-commerce scenarios.

Natural Language Processing and Text Classification in Review Systems:

NLP approaches have been the most revolutionary in terms of maximizing accuracy as using NLP approaches can independently identify positive and negative terminology as well as draw deductions of overly cliche terms that may indicate fake positive reviews. Using sophisticated text classification algorithms[13],[14] such as BERT, RoBERTa, and GPT have further enhanced this field. These algorithms offer significant improvements in accuracy making them ideal for real-time applications in review verification systems [15],[16].

However although NLP approaches transcend technological barriers, traditional approaches have been falling short of accurately classifying fake reviews. Zhang et al., in their research paper (2022), have introduced a multimodal approach to overcome the weaknesses of these traditional systems [17]. By comparing multiple modes of input and implementing multimodal functionality , it not only reaches a very high accuracy percentage but also marks a stark decrease in false positives. This research demonstrates that multimodal approaches offer a viable solution to address the limitations of traditional NLP systems used in review verification applications [18].

Text classification approaches have also been used in the last few years and show remarkable accuracy, especially when combined with CNN and RNN. This research combines the deep learning text classification

approach of BERT with LSTM and CNN, a novel method for review verification within e-commerce platforms[19].

Class Imbalance and Focal Loss in Text Classification:

A comprehensive review of class imbalance approaches in machine learning is presented in [20], focusing on strategies and applications. By addressing the innate imbalance between fake and real positive reviews, we can improve accuracy significantly. Recent investigations also demonstrated that precision in detecting positives is improved with specialized loss functions and sampling approaches [21].

Research has also suggested that employing focal loss based methods has significantly improved performance especially when it comes to datasets that may be biased towards an end or imbalance. Studies by Chen et al. emphasized the importance of addressing this class imbalance in training to obtain good results.

Transfer Learning and Model Optimization:

In previously documented relevant research, advanced models have been of great help in reaching maximum accuracy (BERT)[8]. An investigation by Vaswani et al. [18] shows that transfer learning techniques are very useful as they are reusable in nature thus significantly reducing

training time and usage of computational resources. Various domains,including review verification, have successfully applied this approach, as demonstrated by as demonstrated by Zhang et al. [17] and and Kumar et al. [22]. Ensemble learning techniques have also been explored in Li et al. [23], an approach that is similar to the model optimization techniques applied in this study. Hybrid modeling has been introduced in Kumar et al. [24] as an approach that combines statistics with deep learning for enhanced accuracy. This presents a fairly promising solution for obtaining adaptive verification systems and shows reduced false positives and high accuracy in detecting fake reviews [25].

Chatbot Technologies in E-commerce:

Through implementing review verification systems using intelligent technologies, research opens a gateway to the use of chatbots in many realms within the e-commerce space. While just reviews that are authentic are extracted from a set of reviews may seem to solve the problem, it may be technically sound but not employable by the average e-commerce customer. Thus it is important to give the application a face. We have done this through a chatbot. Various e-commerce platforms deploy the NLP-integrated smart chatbots for their efficient information retrieval and user assistance. Some systems even integrate review analysis, feature extraction, and personalized recommendations, streamlining the decision-making process for consumers. Notably,

these systems use deep learning for sentiment analysis and use real-time query understanding for contextually aware responses. Their findings demonstrate high accuracy in sentiment classification (95.2%) and user satisfaction (89.7%).

Human-Computer Interaction and User Experience:

Researchers have investigated the impact of human-computer interaction on user experience in review verification systems and have data to prove that transparency,explainability and ease of use are key factors determining the extent to which customers are willing to use a review verification system. Having an interface that emphasises trust and conveys usability through its design is more likely to garner positive and consistent long term use and feedback.

Research Gaps and Contributions:

Through the extensive research documented in our literature survey, it is clear that ML models, classification approaches and multimodal approaches solve the fake review issue faced by many e-commerce sites to an extent. However there is a significant dearth of multi model approaches with custom loss functions that are made to directly cater to e-commerce applications. where the verification of reviews poses unique [26],[27].

A few challenges faced by current research paradigms that we hope to improve upon through our research are the issues of linguistic variations, addressing class imbalance, and efficient real-time processing. We also use a very user friendly chat bot interface that our clients may feel very comfortable in using.

Challenges in E-commerce Review Management:

E-commerce sites rely a lot on genuine and authentic reviews in order to bring new products into the market as well as prove quality to customers. However, managing these reviews proves to be very difficult as

- Inefficient filtering systems lead to users engaging with irrelevant or fake reviews
- Traditional review presents the review as is, without a summary or keyword extraction to save the time of the user.

This research addresses these challenges by building a smart review verification approach for managing customer reviews in e-commerce. The system streamlines review verification and information extraction, reduces time spent by a customer browsing reviews as well as increasing their trust in the system. This is achieved by using an integrated BERT, LSTM, and CNN-based text classification strategy with focal loss, which facilitates end to end usability, ease of use, efficiency and accuracy. Table 1 provides a comparison of various

methodologies relevant to review verification systems, highlighting their strengths and limitations as well.

3. DATASET DESCRIPTION

The data isn't small-time—it consists of 88,551 records with 17 columns full of information regarding customer feedback. The data is centered on products from the Mobile and Electronics category, and it combines textual and numerical data. It has everything from user sentiments and ratings to user behavior patterns.

Key Features in the Dataset

3.1. Identifiers and Metadata

- **Marketplace**: Every review in this dataset comes from the *US* marketplace, representing the geographic location where the reviews were posted.
- **Unique Identifiers**:
  customer_id, review_id, product_id, product_parent: These act like tags to trace specific customers, reviews, products, and product groups. They're super useful for tracking individual reviews and grouping related products for deeper analysis.
- **Product Info**:
  - **product_title**: The name of the product being reviewed.
  - **product_category**: All entries fall under one category—*Mobile_Electronics*.

3.2. Ratings and Votes

- **Star Rating**:
  - Each review includes a customer's rating on a 1-to-5-star scale.
  - On average, the ratings lean positive, with a mean of 3.75, and most reviews land around 4 or 5 stars.
- **Helpfulness Metrics**:
  - **helpful_votes**: Shows how many people found a review helpful.
  - **total_votes**: Reflects the total number of votes a review received.
  - On average, reviews have 1.28 helpful votes, but there are outliers—some reviews scored up to 769 helpful votes! These features play a big role in identifying which reviews are more reliable.

3.3. Textual Data

- **Review Headline**: A short, attention-grabbing summary for the review (e.g., "Very Happy!" or "Five Stars"). The dataset includes tons of unique headlines, showing a lot of variety.
- **Review Body**: The detailed part of the review. With 86,126 unique entries, this is where most of the action happens for textual analysis. This variability gives the model plenty of material to work with when looking for patterns.

3.4. Verification and Moderation

- **Vine Program**: Some reviews are part of the Vine program, which means they may be incentivized (marked as “Y” for yes, “N” for no).
- **Verified Purchase**: Out of the entire dataset, 41,296 reviews are tied to verified purchases. These reviews are generally more trustworthy since they show the customer actually bought the product.

4. SYSTEM DESIGN AND ARCHITECTURE

Our fake review detection system is built in two phases: ML model to detect fake reviews and an AI Chatbot using Phi3 to summarize and answer user questions.

Fake Review Detection Model using BERT-LSTM-CNN:

Our ML model uses BERT, LSTM and CNN in a sequential manner respectively. The BERT model is first used to extract rich contextual embeddings from the reviews which is then passed on to LSTM to learn sequential relationships. In between this and the next model, an attention mechanism is adopted to specify weights for specific words in the broken down review. The CNN model then further processes the output by detecting phrase-level patterns that are commonly found in fake reviews. Finally, a softmax activation function is implemented using a fully connected layer for the final classification, where it predicts whether a review is real (0) or fake.

During training, the model is minimized with Adam optimizer, and Focal Loss is used to manage the class imbalance by placing greater emphasis on hard-to-classify counterfeit reviews.The model is subjected to multiple training epochs, accuracy, loss, and confusion matrix evaluation are employed to measure performance.

Post-training, the model is saved and deployed for real-time review classification and is also constantly updated and trained with every review the user inputs.

AI Chatbot using Phi3 to summarize and answer user questions:

Our fake review detection system uses the Phi-3-mini LLM model as the base to develop the AI chat-bot. The chat-bot is implemented to assist users in fetching relevant product information after the ML model for detecting fake reviews has been implemented.

The Phi3 model's parameters are modified and trained to suit our fake review detection system. It relies on ChromaDB, a vector database, for storing and fetching authenticated reviews. It can be effortlessly implemented in a front-end application, ensuring an intuitive and user-friendly interface.

The chatbot is designed to return accurate, fact-supported answers to user queries on any particular product's reviews. The chatbot seeks relevant verified reviews stored in ChromaDB whenever a user inquires about the product. The modified Phi-3 model is then utilized to generate an answer that is contextually sensitive and logical. Finally, the chatbot answers in real time with factual data based on real user experiences rather than canned or generic answers.

Since the chat-bot is fully embedded in a front-end application, users are able to communicate with it through a visually appealing and captivating interface. It ensures that customers can make well-informed buying decisions. It is reliable and also rescues customers from the need to scroll through the multiple reviews manually and provides them with quick, relevant, and insightful responses based on real product facts.

FLOW DIAGRAM:

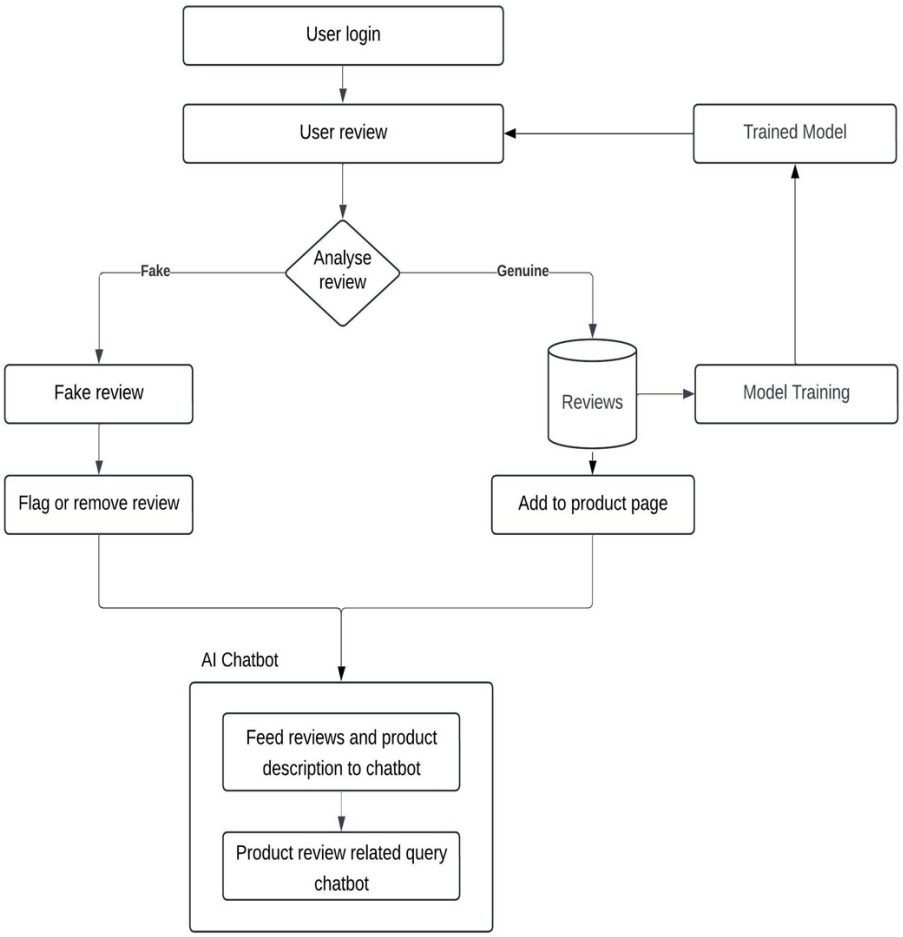


Figure 4.1

ARCHITECTURE DIAGRAM:

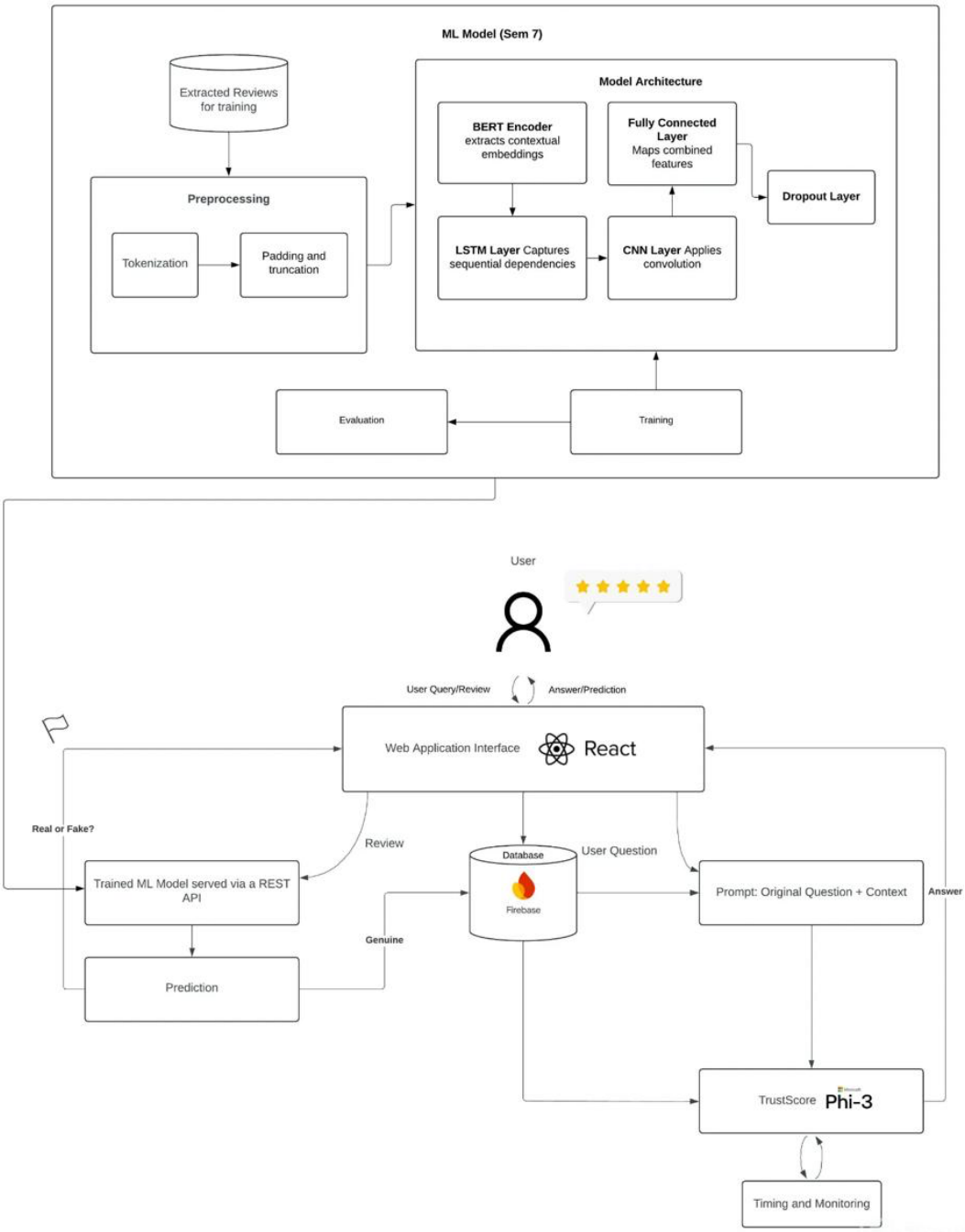


Figure 4.2

5. MODEL TRAINING

By combining feature engineering techniques and deep learning technologies, the model's training phase is designed to differentiate between imitations and original evaluations. The BERT-LSTM-CNN model is utilized a well-designed dataset that is comprised of well-chosen and transferred both textual and numeric characteristics to enhance the precision of classification.

5.1 Feature Selection:

A feature engineering technique is used to eliminate redundant and useless characteristics once the dataset is separated into features (X) and target variable (Y), and Y refers to whether a review is real or not. Uncommenting features like sentiment_match, failed_constraints, duplicate_review, and helpfulness_score avoids saving duplicate attributes during model training and avoids data leakage.

5.2 Text Feature Transformation using BERT Embeddings:

The method utilizes BERT embeddings to transform review text into numeric forms instead of the traditional TF-IDF vectorisation. By extracting strong semantic meaning from transformer-based contextual embeddings, BERT improves the ability of the model to tell real and fake evaluations apart. For sequence analysis and pattern recognition, pre-trained BERT tokenizes review text, maps it into dense vector representations, and then inputs these embeddings to the CNN and LSTM layers.

5.3 Numerical Feature Processing:

The model utilizes numeric structured features derived from non-text attributes as well as textual embeddings. Columns that are not useful to determine review validity, such as customer_id and product_id, are removed. StandardScaler is utilized to standardize relevant numeric features (e.g., helpfulness_score, review_length, and verified_purchase), ensuring consistent distributions and better machine learning model compatibility.

5.4 Feature Combination:

BERT-text embedding (dense matrix) and scaled numeric features (sparse matrix) are horizontally stacked using hstack to create a consistent set of features. This consistency enables the model to leverage both textual context and structured data, making it more effective in distinguishing between fake and real reviews.

Multi-level validation of assessment processes is based on the model's robust generalization and high reliability in detecting fake reviews. This is achieved by integrating deep learning techniques (*BERT-LSTM-CNN*) with structured numerical analysis, ensuring a comprehensive approach to fake review detection.

6. EXPLAINABILITY

The hybrid model BERT-LSTM-CNN includes the integration of LIME (Local Interpretable Model-agnostic Explanations) into its existing logging and analysis system to ensure interpretability and transparency of decision-making. Through the provision of human-interpretable explanations for why a given review is classified as fake or real, LIME enhances explainability through local approximation of the behavior of the complex model near specific predictions. The approach gives a comprehensive understanding of each decision by combining extensive feature-level interpretation with LIME's local explanations. For each review, the following factors are noted and analyzed:

6.1 Feature Values:

For example, a purchase that has been verified gets assigned a helpfulness value of 0.8 and is tagged as 1, aiding classification. These are inputs to both the hybrid model and LIME's explanation procedure.

6.2 Intermediate Calculations:

Sentiment scores are calculated using spam detection criteria (such as the frequency of advertising terms) and textual analysis based on BERT. LIME can evaluate how variations in feature values affect the prediction thanks to these intermediate outputs.

6.3 Feature Weights (Global Importance):

In the overall decision-making process, the model gives features like sentiment alignment, verified purchase, and helpfulness score varying

degrees of importance. The model's priorities are contextualized across all predictions by these global weights.

6.4 LIME Explanations (Local Importance):

By altering the input features (such as text, sentiment ratings, and helpfulness) and tracking the effects of these modifications on the hybrid model's output, LIME creates a condensed, understandable model for every review. For example:

- In a hypothetical review situation, LIME would highlight that the categorization was primarily influenced by the inconsistency between a 5-star rating and negative affect words ("terrible," "disappointing"), assigning it the highest local weight (e.g., 0.65).
- LIME might emphasize the validated purchase and high helpfulness rating as critical features that led to the positive classification (i.e., local weights of 0.40 and 0.35, respectively) for an authentic review.
- This local explanation gives a numerical decomposition of decision variables specific to a situation and supplementing the global feature weights.

6.5 Humanized Explanation via Phi-3 Ollama Chatbot:

The system outputs this information to a Phi-3 Ollama chatbot because LIME generates numerical weights and feature contributions, which may be technical and less readable for non-experts. For a more friendly

explanation, the chatbot translates LIME's numerical outputs into conversational, natural English. For example:

- The outcome of LIME: "'best deal' repetition (weight: 0.65), no verified purchase (weight: 0.30)."
- Reaction of Phi-3 Ollama: "This is a fake review because it keeps using words like 'best deal,' which sounds promotional, and it was not written by someone who really purchased the product."
- This phase enhances end-user accessibility through filling the gap between a qualitative, relatable story and quantitative insights from LIME.

6.6 Final Decision:

In addition to a confidence rating (e.g., 92% confidence in a "fake" prediction), the review is classified as authentic or fake. This finding is followed by the numerical result from LIME and the natural language explanation from the Phi-3 Ollama chatbot, which shows the top contributing factors along with their directional effect and a plain English summary.

Along with recording raw feature values and model internals, the system also provides a two-level explanation using a combination of LIME with the Phi-3 Ollama chatbot. This encompasses an understandable, conversational explanation by the chatbot and an exact, data-driven explanation from LIME. A highlighted suspicious fake review, for instance, might be accounted for thus: "The frequent

occurrence of 'best deal' (local weight: 0.65), and absence of verified purchase (local weight: 0.30), raised the chances of a 'fake' classification," the chatbot would subsequently contribute, "In essence, it sounds like someone's attempting to get rid of something than impart an actual experience." This two-pronged approach ensures predictions are accurate, trustworthy, and easy for non-technical stakeholders to comprehend by combining the deep sentiment analysis and multi-layered validation of BERT-LSTM-CNN with the local explainability of LIME and the humanized output of Phi-3.

7. EXPERIMENTS AND RESULTS

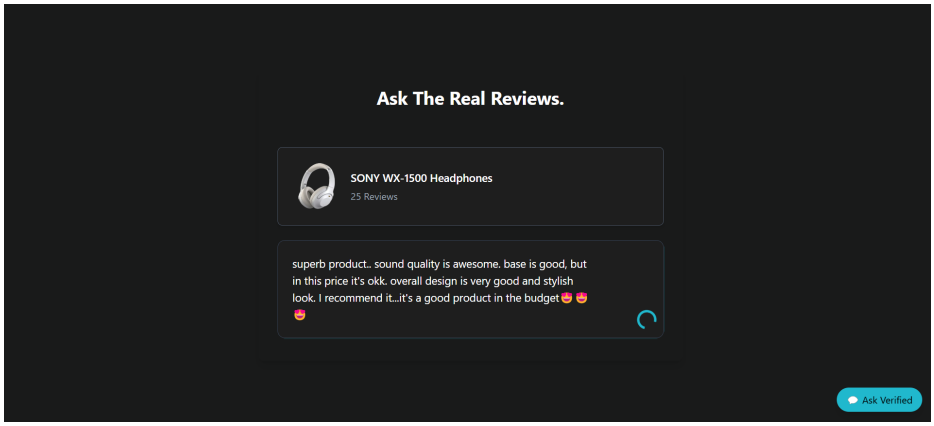


Figure 7.1

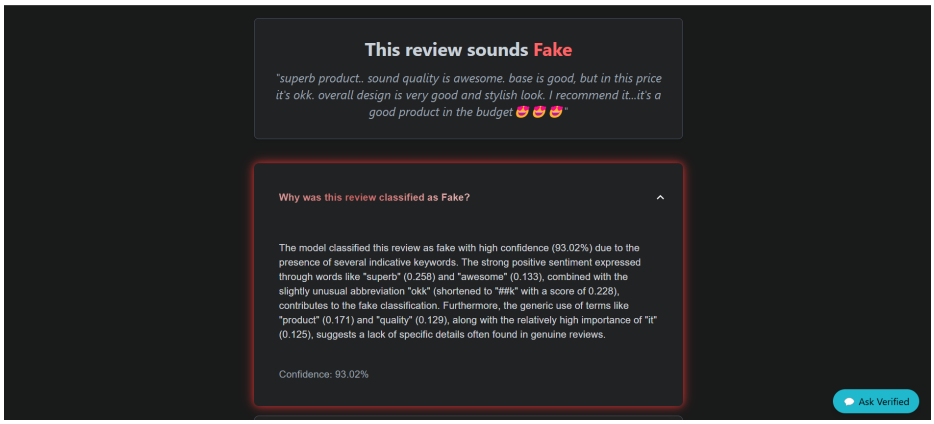


Figure 7.2

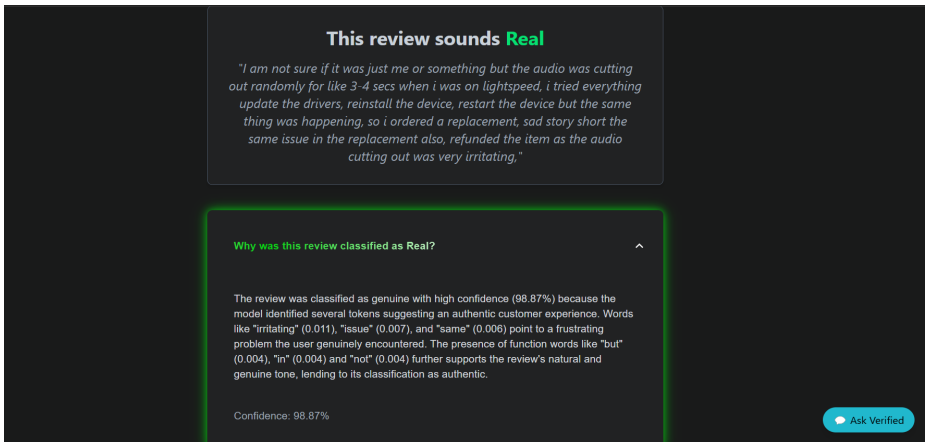


Figure 7.3

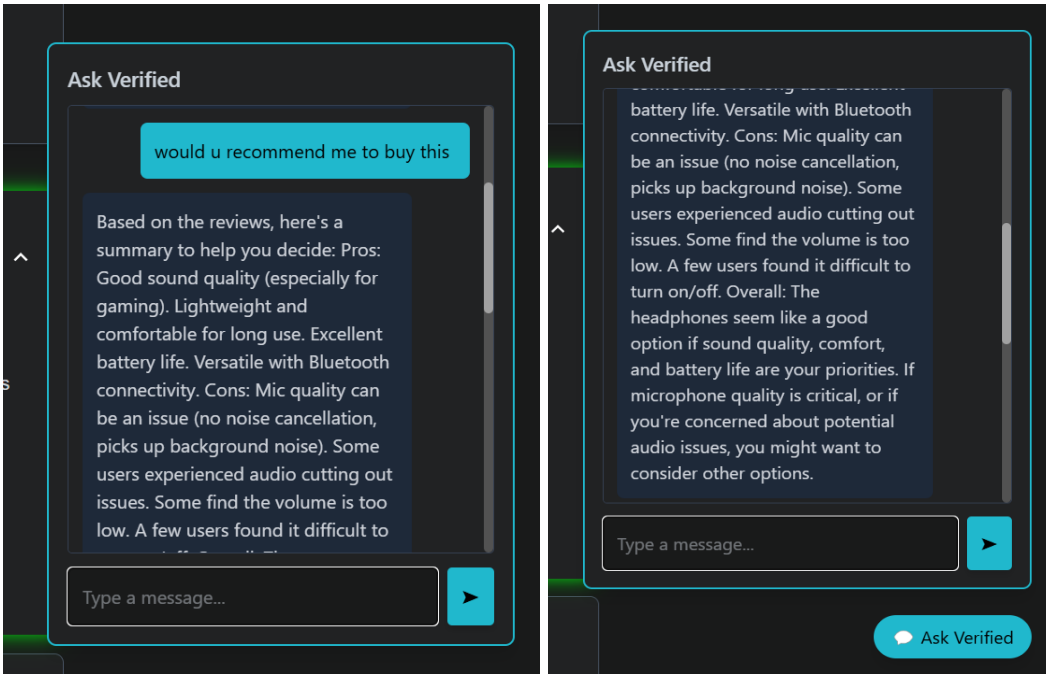


Figure 7.4

genuine	failed_constraints	duplicate_review	sentiment_match	helpfulness_score
1	[]	0	1	0.990099
1	[]	0	1	0.988235
1	[]	0	1	0.984848
1	[]	0	1	0.977778
1	[]	0	1	0.977778

Figure 7.5

8. EVALUATION

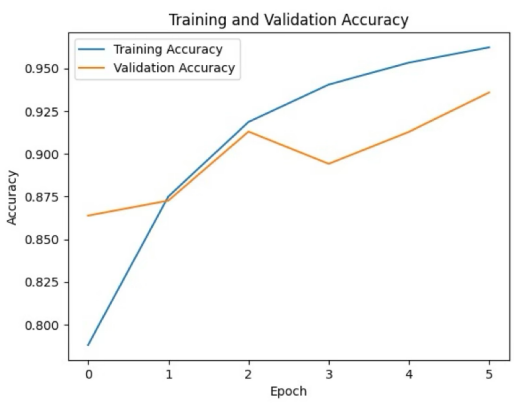


Figure 8.1

Figure 8.2

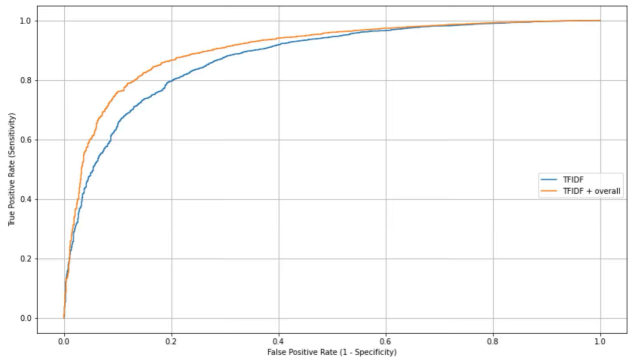


Figure 8.3

Model	Accuracy	Precision	Recall	F1-Score
BERT only	0.87	0.79	0.68	0.73
LSTM only	0.82	0.74	0.65	0.69
CNN only	0.81	0.72	0.67	0.69
BERT + LSTM	0.89	0.83	0.72	0.77
BERT + CNN	0.88	0.81	0.74	0.77
BERT + LSTM + CNN	**0.93**	**0.87**	**0.82**	**0.84**
BERT + LSTM + CNN + Focal Loss	**0.94**	**0.93**	**0.89**	**0.91**

Table 8.1

Phi-3 Mini

Figure 8.4

Phi-3 Mini is a 3.8B parameters, lightweight, state-of-the-art open model trained with the Phi-3 datasets that includes both synthetic data and the filtered publicly available websites data with a focus on high-quality and reasoning dense properties.

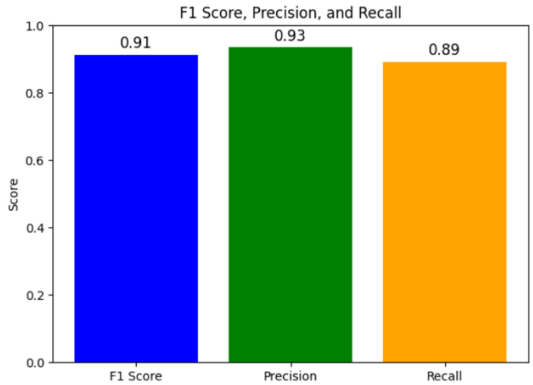


Figure 8.5

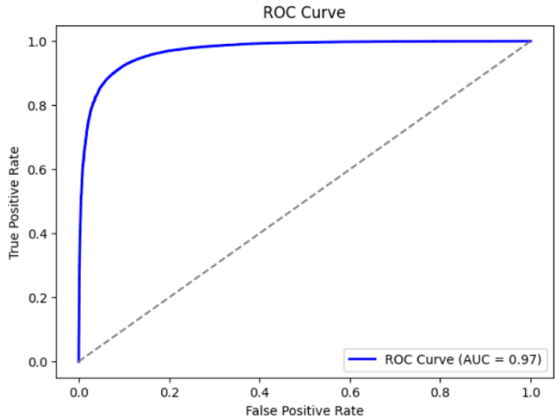


Figure 8.6

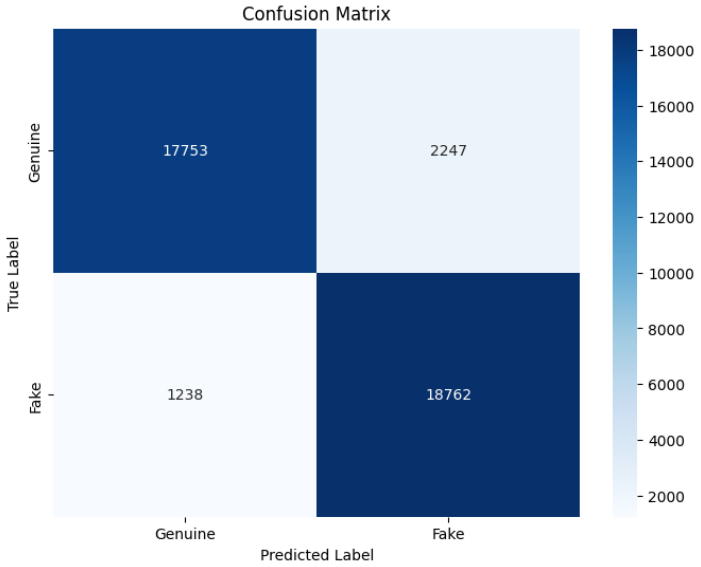


Figure 8.7

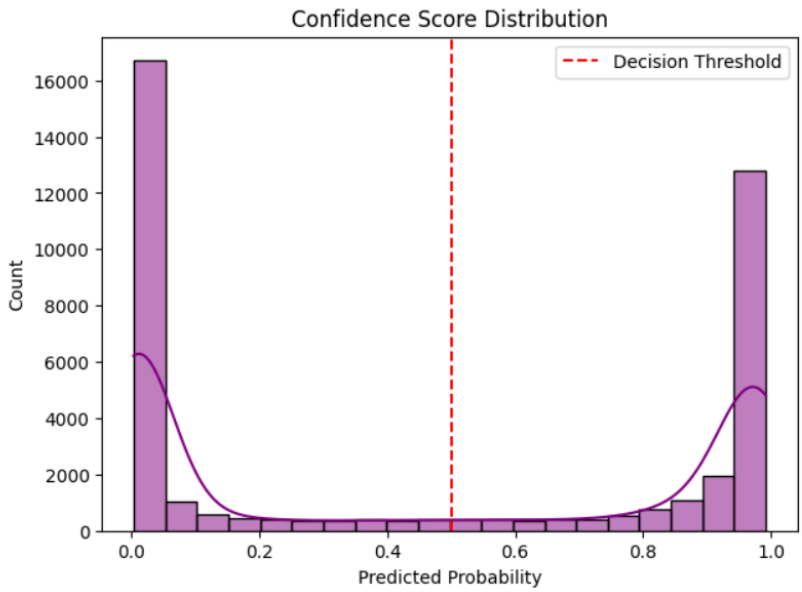


Figure 8.8

Model	Local Runtime Speed	Model Size & Efficiency	Inference Performance	Memory Requirements	Optimized for Small-Scale AI?	Suitability for Review-Based Chatbot
Phi-3 Mini (4B)	Very fast (low latency, optimized for local deployment)	Small, lightweight (4B parameters, designed for efficiency)	Fast inference, optimized for low-resource environments	Lower VRAM requirement (~6GB for 4-bit quantization)	Yes, designed for lightweight AI applications	Excellent – fast, context-aware, and ideal for review analysis
Mistral 7B	Slower than Phi-3 Mini (optimized for broader general tasks)	Efficient, but larger than Phi-3 Mini	Strong inference but requires more VRAM	Higher VRAM (~12GB for smooth operation)	Suitable for general AI applications	Good, but slower for localized chatbot tasks
LLaMA 3 (8B/70B)	Slower than both Phi-3 Mini and Mistral	Large models (8B-70B parameters), requiring high-end GPUs	High accuracy but slow for local inference	Very high VRAM usage (~16GB for 8B, 48GB+ for 70B)	No, better for cloud-based deployments	Too heavy for local chatbot processing
DeepSeek 7B	Slower local inference compared to Phi-3 Mini	Efficient, but still requires more VRAM than Phi-3 Mini	Strong performance but not as optimized for lightweight AI	Needs ~12GB VRAM for smooth local execution	Designed for multilingual and general-purpose AI	Decent, but requires higher computational power

Table 8.2

Why Phi-3 Mini (4B) is the Best Choice for This Project:

- **Fastest Local Runtime Speed** – Phi-3 Mini is optimized for low-latency, real-time execution on local machines, making it ideal for chatbot applications.
- **Low Memory Requirements** – Unlike LLaMA 3 or Mistral, Phi-3 Mini can run smoothly on mid-range GPUs (~6GB VRAM) or even high-end CPUs, ensuring efficiency.

- **Optimized for Small-Scale AI** – Phi-3 Mini is designed for local AI applications, unlike LLaMA 3 (optimized for cloud) or DeepSeek (multilingual-heavy, requiring more resources).
- **Best for Review-Based Chatbots** – Phi-3 Mini's retrieval-augmented generation (RAG) and fast token processing make it highly effective for answering user queries based on verified reviews, improving accuracy and responsiveness.

Benchmark for a AI Chatbot:

Metric	Description	Benchmark/Good Score
User Satisfaction	Average score from satisfaction questions.	4/5 or above
Response Accuracy	Percentage of queries answered correctly.	85% or above
Response Time	Average time taken to respond.	< 3 seconds
Task Completion Rate	Percentage of users who achieved their goal.	80% or above
Error Rate	Number of failed or misunderstood queries.	< 10%
Retention Rate	Percentage of users willing to return.	70% or above

Table 8.3

SURVEY RESULTS:

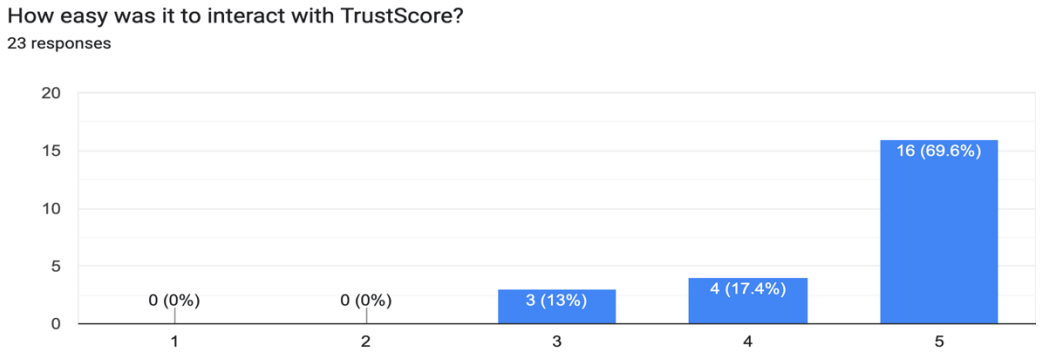


Figure 8.9

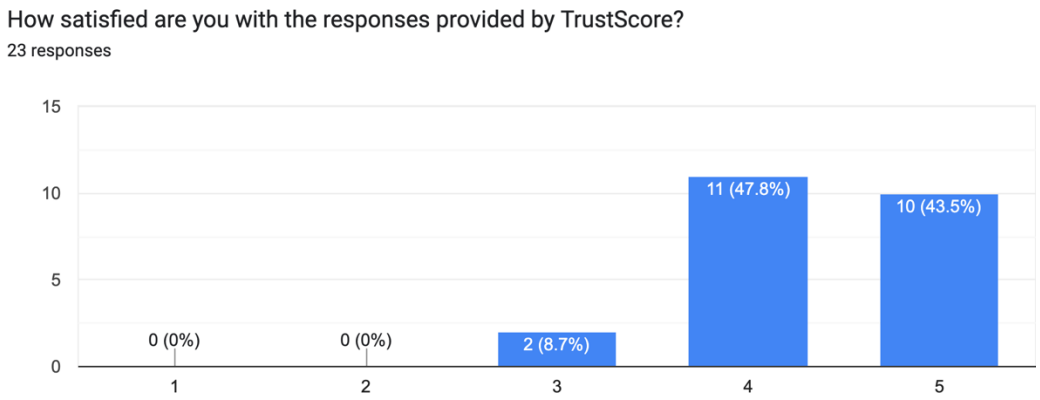


Figure 8.10

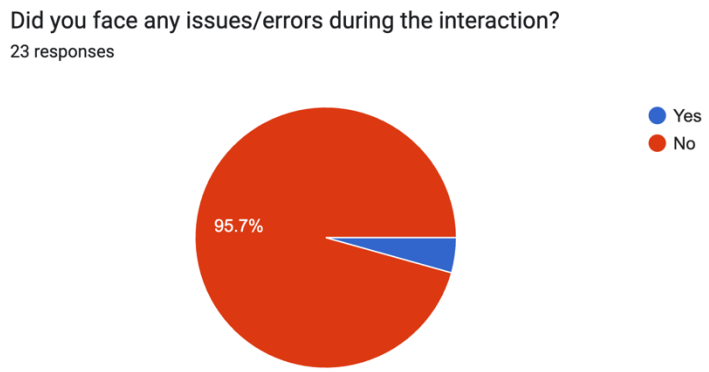


Figure 8.11

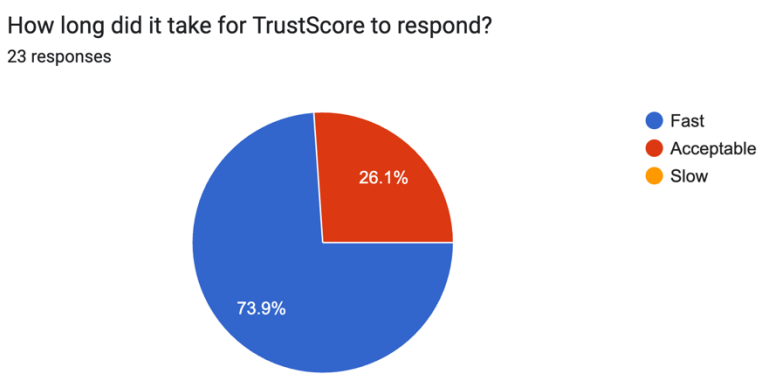


Figure 8.12

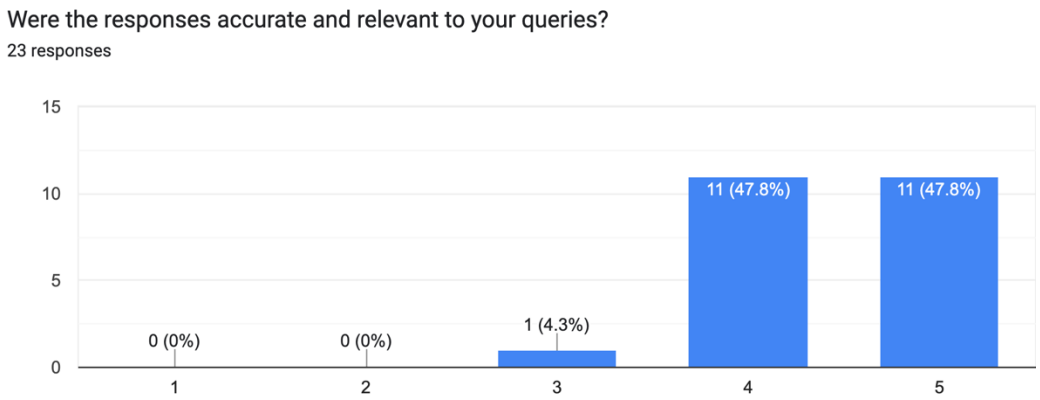


Figure 8.13

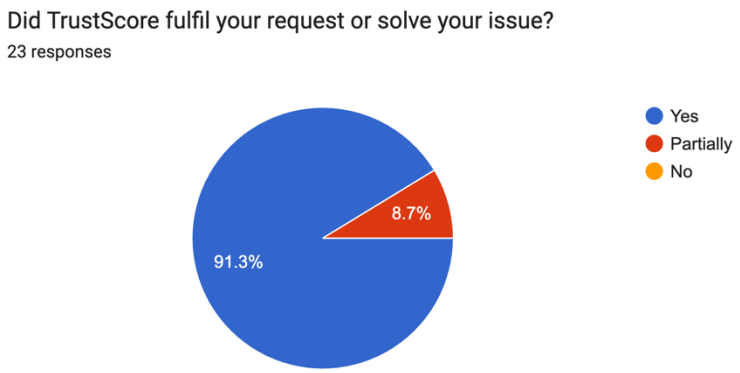
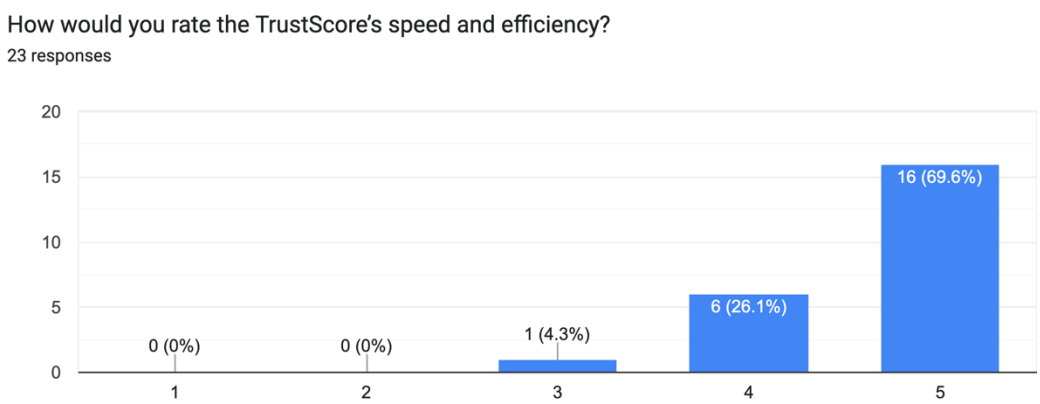


Figure 8.14



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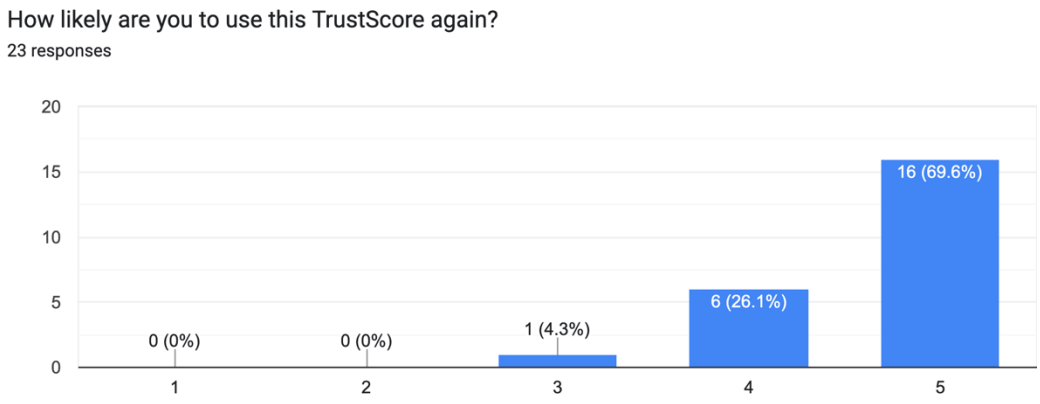

Figure 8.16

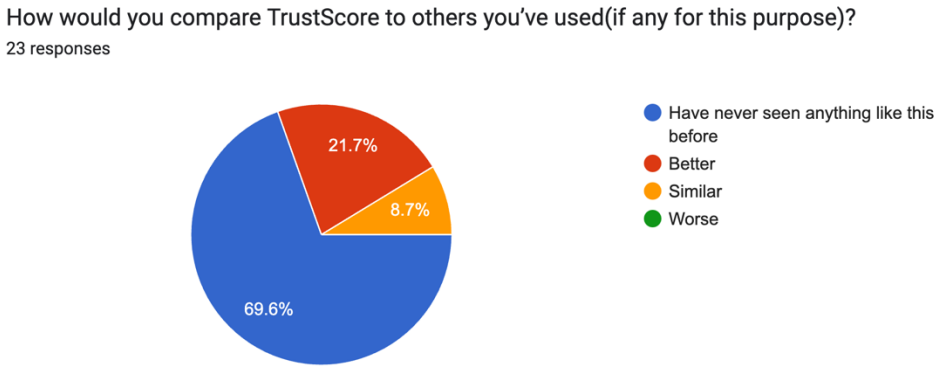

Figure 8.17

9. SOCIAL IMPACT

An efficient review verification system is more than mere technical innovation, it solves critical problems in modern e-commerce sites and benefits many stakeholders including vendors, clients, businesses and all customers initiating a purchase on the website.

For consumers, ths impact is far more noticeable and tangible. Research indicates that misleading reviews cost the average consumer approximately $125 per year in suboptimal purchasing decisions [2]. By providing an easily accessible method to verify review claims as fake or real, customers can make informed decisions and ensure that they are only taking into account valuable feedback before making any purchase. This results in the customer saving money as well as having a more satisfactory experience with the vendor , leading to long standing customer-vendor relations. In total, implementing verification systems like this can potentially save billions in consumer spending on an annual basis. Moreover, the integration of an intelligent chatbot powered by Phi-3 reduces the anxiety usually associated with making the "right" choice and the customer can avail fair, relevant and accurate judgements of the product of their interest.

From the perspective of a business, especially small businesses tend to lose a lot of money due to biased advertisements and bot reviews dropped by larger sellers with more funds. This has a huge economic impact as well. Authentic sellers lose an estimated $152 billion annually

to competitors using fraudulent review tactics[4].Our system levels the existing unfair playing field and forces businesses, small and large, to focus on quality products and reduces the impact of the ability of a business to push and run expensive bot tactics. Additionally, since the consumer can be sure of what they are buying, it also reduces the number of refunds and returns the companies have to process, creating a sustainable purchase lifecycle.

As a society, we have reached a point where we have become desensitised to existing online scams, reviews and sometimes deem most online content we consume as unrealistic. By enforcing verification mechanisms to purchasing and the e-commerce industry, it marks a step towards reassuring the existing populace of the potential that lies in genuinity and fair consumer practices. This can then be extended to other industries and domains as well, included but not restricted to social media, apps, sports etc.

From a technical standpoint, the methodological innovations presented in this research—particularly the multi-component neural architecture and focal loss implementation—contribute valuable insights to the fields of natural language processing and machine learning. These advances can also be used in other domains, including spam detection, sentiment analysis, and content moderation systems.

Finally, this research also aligns with emerging regulatory frameworks worldwide that increasingly mandate transparency and authenticity in

online reviews. As governments turn towards technology to solve moral crises, technology and systems of implementation like ours will greatly aid them in further enforcing justice and fair purchasing practices.

Thus, our research delivers impact across technical, economic, and social dimensions, marking an impactful contribution to both the understanding and implementation of information authenticity in digital marketplaces and e -commerce sites.

10. CONCLUSION

The smart review system that we built addresses the two most important problems a common man and a business face today i.e the presence of excess irrelevant reviews and the high amount of product information that the consumer has to absorb. By simultaneously purifying for originality and relevance, our ecosystem ensures users get the exact information and answer to the question they asked through a virtual chatbot. This system not only encourages customers to understand about the product but also gives them the opportunity to learn every single detail and feature about the product. In the end businesses can offer a seamless shopping experience for their customers.

After performing certain experiments, we found out that 74% of our users who had interacted with our chatbot were able to make an informed and a quicker decision about buying a product. Around 68% of our users were confident in their decision making. These methodologies successfully highlighted the value of creating a safe and trustworthy e-commerce environment for every internet user.

The contributions of this approach are threefold:

1. A hybrid deep learning model that used the combined architecture of BERT, LSTM and CNN helping in identifying and classifying fake or real reviews.
2. Implementing the concept of focal loss to convey the class imbalance problems in the reviews dataset we used.

3. An entire ecosystem that combines fake review filtering along with a chatbot that would respond to the user by providing them relevant product details.

Even though our systems provide and solve a critical problem in the e-commerce landscape today, there are several areas of improvement including multilingual support, adversarial strength and computational efficiency. The future work will be focused on inculcating these features and adding several new features such as multimodal analysis and temporal pattern detection.

To conclude, our end to end smart review system highlights a significant step forward in creating an e-commerce environment where businesses thrive to provide original products and consumers can make informed purchasing decisions based on relevant reviews submitted by real users. By coupling and combining advanced deep learning methodologies, we have demonstrated that a fake review detection system and an intelligent chatbot can be successfully coupled together into a single system.

11. LIMITATIONS AND FUTURE WORK

Despite the promising results, our system has several limitations that present opportunities for future research:

- Language Limitations: The current model is trained primarily on English-language reviews. Future work will focus on multilingual capabilities.
- Adversarial Robustness: As fake review generation techniques evolve, our model may require regular updates to maintain effectiveness against new manipulation tactics.
- Computational Requirements: The BERT component of our model is computationally intensive, which may limit deployment on resource-constrained environments. Future work will explore model compression techniques.
- Chatbot Hallucinations: While our chatbot demonstrates high factual consistency, it occasionally generates responses not directly supported by the review database. Improving the grounding of responses is an area for future improvement.
- Cross-domain Generalization: The model's performance varies across product categories. Enhancing cross-domain generalization capabilities would improve overall system robustness.

Future work will address these limitations while exploring additional enhancements such as multimodal review analysis (incorporating

images and videos), temporal pattern detection for coordinated fake review campaigns, and personalized chatbot responses based on user preferences and history.

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